

LES GRANDES CHUTES, QC, Canada

WMO: 713910

Lat: 45.840N

Lon: 75.649W

Elev: 638

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.5	(c) -10.7	(d) -23.0	(e) 1.6	(f) -14.6	(g) -18.1	(h) 2.1	(i) -9.4	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.9	(c) 85.2	(d) 70.7	(e) 81.9	(f) 68.4	(g) 79.0	(h) 67.0	(i) 73.3	(j) 80.5	(k) 71.4	(l) 78.2	(m) 69.6	(n) 75.8	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	117.3	76.3	69.3	110.1	74.6	67.5	103.6	72.8	37.3	80.7	35.7	78.5	34.1	76.0	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) -21.1	(f) 89.7	(g) 7.1	(h) 3.1	(i) -26.3	(j) 92.0	(k) -30.4	(l) 93.8	(m) -34.4	(n) 95.5	(o) -39.6	(p) 97.8		
(4) N/A	(5) N/A	(6) N/A	(e) -21.3	(f) 76.7	(g) 7.2	(h) 2.6	(i) -26.5	(j) 78.6	(k) -30.8	(l) 80.2	(m) -34.9	(n) 81.6	(o) -40.1	(p) 83.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.9	12.1	14.9	25.8	39.9	53.6	62.8	67.5	66.1	58.6	45.6	33.4	20.3		
	DBStd	21.32	13.07	11.35	10.96	8.23	7.48	6.34	5.06	5.17	7.07	7.83	8.41	11.27		
	HDD50	4884	1174	982	751	320	46	0	0	0	9	183	500	919		
	HDD65	8715	1639	1402	1215	754	361	115	30	50	215	601	949	1384		
	CDD50	1918	0	0	1	16	159	385	544	498	266	48	1	0		
	CDD65	273	0	0	0	0	9	49	109	83	22	1	0	0		
	CDH74	2101	0	0	0	7	135	461	818	533	145	2	0	0		
	CDH80	494	0	0	0	1	30	118	206	109	30	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	39.0	2.8	2.3	2.5	2.9	3.0	3.7	3.8	3.8	3.8	3.5	3.3	3.3		
	PrecMax	48.7	4.3	4.8	4.9	5.2	5.1	6.7	7.0	5.5	6.5	6.2	5.9	5.6		
	PrecMin	33.6	1.2	0.9	1.0	0.9	1.3	1.1	1.2	1.7	1.5	1.1	1.2	0.9		
	PrecStd	3.6	0.8	1.0	1.1	1.2	1.1	1.4	1.6	0.9	1.4	1.3	1.1	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.3	43.0	55.9	73.0	84.3	88.7	89.6	87.5	84.2	72.0	57.3	47.1		
		MCWB	41.5	39.5	45.5	58.1	67.1	72.4	74.7	70.5	71.3	62.8	52.9	45.8		
	2%	DB	37.4	37.7	48.4	64.9	77.7	83.4	85.2	82.9	77.8	65.9	52.8	41.1		
		MCWB	35.9	35.0	41.1	52.3	61.6	68.6	71.4	69.3	68.2	58.7	49.3	39.9		
	5%	DB	34.0	34.8	44.1	58.9	72.4	79.4	81.8	79.6	73.8	61.4	48.6	37.1		
		MCWB	32.9	32.9	37.9	48.1	58.1	66.1	69.0	67.9	65.4	55.7	45.6	35.8		
	10%	DB	31.1	32.4	40.2	53.7	67.8	75.5	78.6	76.6	70.5	57.4	45.1	34.2		
		MCWB	29.8	30.6	35.6	44.5	55.9	64.0	67.6	66.4	64.0	53.3	42.4	32.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.8	39.8	49.2	60.1	70.1	75.3	77.1	74.3	73.0	64.7	54.9	46.1		
		MCDB	41.9	41.5	53.1	68.8	80.5	84.4	86.5	80.7	79.5	68.9	56.3	46.6		
	2%	WB	36.3	35.5	42.0	55.0	65.3	71.6	73.7	72.3	70.1	60.1	49.8	40.2		
		MCDB	37.1	37.1	46.6	61.3	72.9	80.4	81.0	78.8	75.0	64.0	51.7	41.2		
	5%	WB	33.1	33.2	38.7	49.6	61.6	69.0	71.6	70.4	67.6	56.7	46.2	35.9		
		MCDB	34.0	34.5	42.9	56.3	67.4	75.7	78.1	76.2	72.3	60.4	48.2	36.8		
	10%	WB	29.9	30.6	36.2	46.0	58.3	66.4	69.6	68.5	65.1	53.7	42.6	33.1		
		MCDB	31.1	32.5	39.5	52.7	64.9	72.1	75.8	74.0	69.4	57.2	44.8	33.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	18.8	18.4	18.3	19.7	19.1	18.9	18.3	17.7	14.1	11.9	12.5		
		MCDBR	14.7	16.4	21.0	28.9	28.8	25.8	23.4	22.4	21.8	20.3	17.0	12.1		
		MCWBR	14.2	14.5	14.4	16.7	13.7	12.6	11.2	10.5	12.6	13.2	14.0	11.9		
	5% WB	MCDBR	14.7	15.4	18.8	24.9	22.8	21.3	19.9	18.7	19.4	18.6	16.0	12.1		
		MCWBR	14.5	14.3	14.1	16.7	13.1	12.4	11.1	10.0	12.5	13.5	14.3	12.3		
(40) Clear-Sky Solar Irradiance	taub		0.255	0.280	0.321	0.337	0.374	0.401	0.403	0.387	0.363	0.331	0.305	0.272		
	taud		2.363	2.273	2.302	2.427	2.369	2.305	2.307	2.364	2.417	2.509	2.500	2.369		
	Ebn at Noon		274	285	288	293	283	275	272	273	271	265	253	252		
	Edh at Noon		23	31	35	34	37	40	40	36	31	25	21	21		
(44) All-Sky Solar Radiation	RadAvg		433	744	1172	1442	1634	1785	1793	1580	1230	720	433	316		
	RadStd		58	82	105	134	158	134	155	77	105	67	52	27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page